CG CRASH GUARD

SEE THE INCIDENT.
MOVE THE RESPONSE.

AI-powered traffic accident detection, evidence capture, and authority response.



CAMERA PANEL + AUTHORITY MANAGEMENT

FINAL YEAR PROJECT / SYSTEM WALKTHROUGH

One incident. One connected workflow.

Camera-side detection preserves evidence and shares incident details with the authority workflow.

01

CAMERA FEED

Traffic video enters the monitoring node.

02

YOLOv8

The model checks frames for accident classes.

03

EVIDENCE

Pre-event buffer plus post-event recording.

04

AUTHORITY

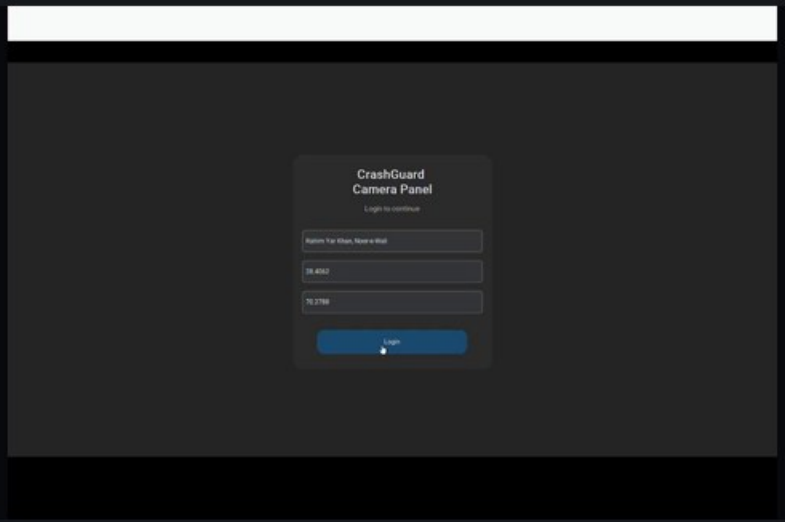
Incident record, video, location, and alert.

10s PRE-EVENT BUFFER

10s POST-EVENT CLIP

Configure the camera node.

Each monitoring node is identified by its location and coordinates.



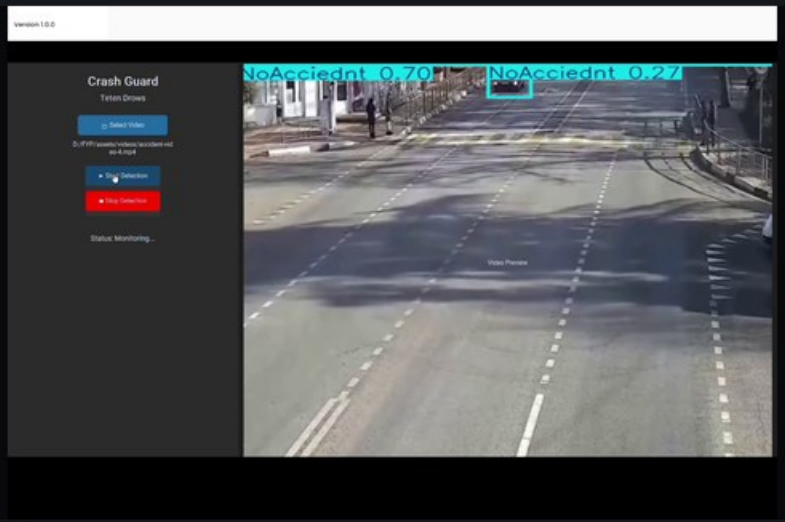
CAMERA SETUP

Enter the camera address and its latitude / longitude before monitoring.

Recorded camera-panel demo

AI monitoring on the video feed.

The desktop panel displays model detections over the selected traffic video.



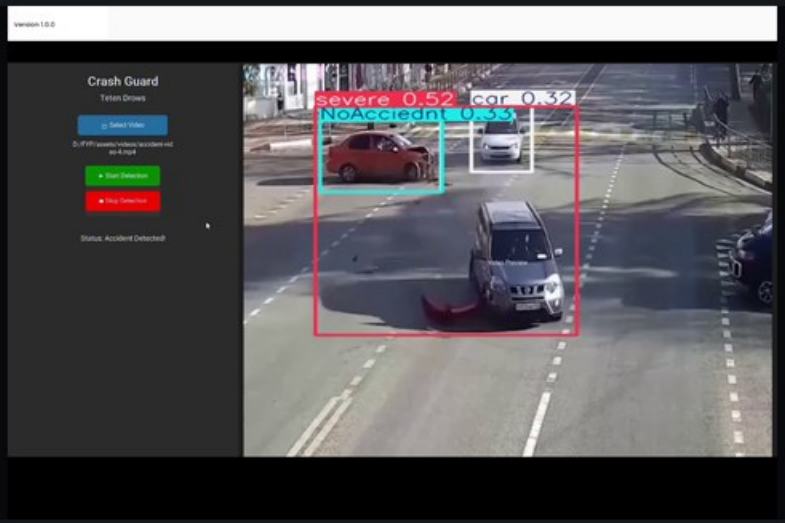
LIVE DETECTION

YOLOv8 labels vehicles and accident severity classes in the operator preview.

Recorded demo footage

Preserve context around an incident.

A rolling frame buffer keeps recent footage available when a detection triggers.



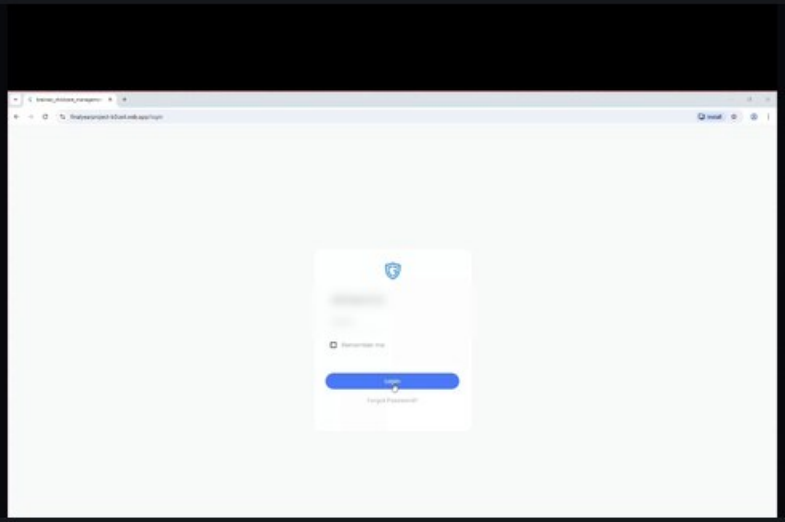
VIDEO EVIDENCE

Capture pre-event context, then continue recording after the trigger.

Recorded demo footage

Authority sign-in screen.

The recording includes a responder sign-in view for the management panel.



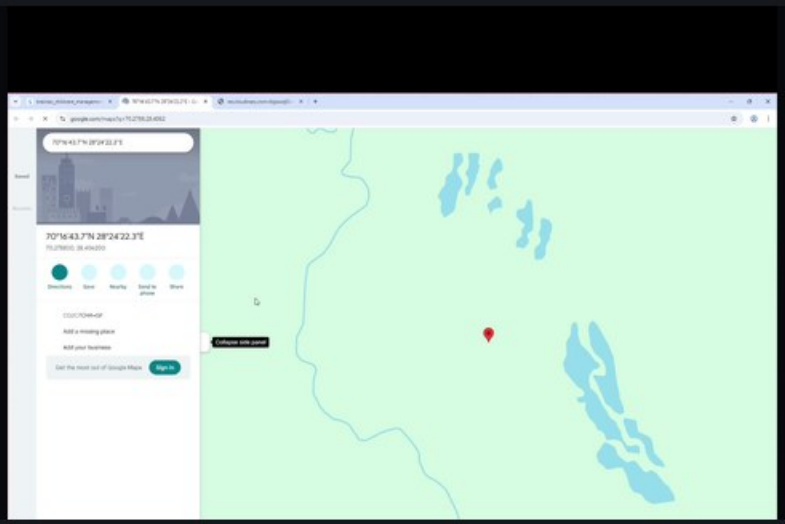
AUTHORITY ACCESS

Sign-in screen shown in demo footage. Visible credential fields are blurred.

Recorded authority-panel demo · credentials obscured

Authorities review the reported location.

The management-panel demo includes a map view alongside incident information.



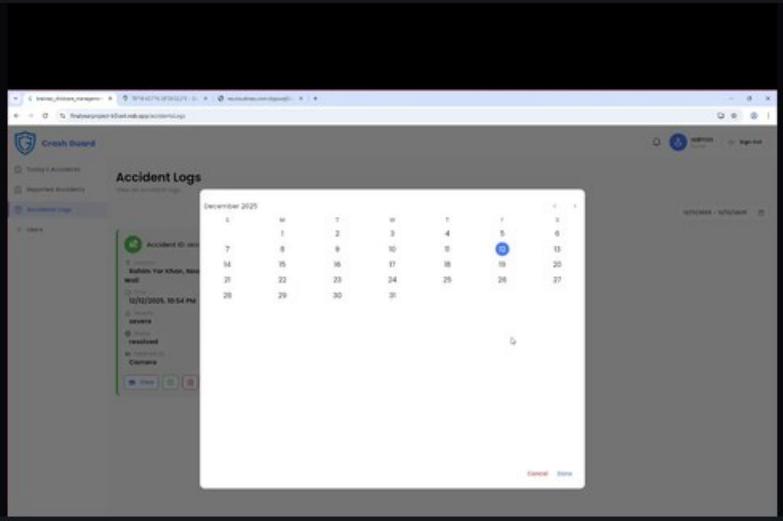
AUTHORITY VIEW

Use the incident location to support review and coordination.

Recorded demo footage

Review earlier accident reports.

The recorded management-panel demo also shows an accident-log view.



INCIDENT HISTORY

Open a report to inspect its recorded details and evidence.

Recorded demo footage

Detection to dispatch-ready information.

The upload worker handles cloud operations separately from the video preview loop.

DETECT
YOLOv8

CAPTURE
Pre + post-event

PUBLISH
Cloud services

REVIEW
Authority panel

CLOUDINARY
Evidence video

FIRESTORE
Incident record

FCM
Responder alert

Cloud video · Firestore metadata · Firebase Cloud Messaging

CRASH GUARD

AI-assisted accident detection and evidence delivery for faster incident awareness.

DETECT / **PRESERVE** / **NOTIFY** / **RESPOND**

CAMERA PANEL + AUTHORITY MANAGEMENT